

# Lecture 05

## Application and Impacts of Computing

Chapter lecture — 12 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XI · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — گیارہویں جماعت

12 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

# How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی ای کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

## Lectures in this pack

Chapter 5: Application and Impacts of Computing. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

**L01.** 5.1 Introduction to Computing Impacts

**L02.** 5.1.1 Artificial Intelligence (AI)

**L03.** 5.1.2 Internet of Things (IoT) & Components

**L04.** 5.1.3 Data Analytics & Components

**L05.** Table: IoT vs AI vs Data Analytics Comparison

**L06.** 5.2 Uses of IoT in Different Areas

**L07.** 5.2.1 Scope of AI in Education in Pakistan

**L08.** 5.3 Information Sources (Primary, Secondary, Tertiary)

**L09.** 5.4 Impacts of Computing in Various Fields

**L10.** 5.5 Assistive Technologies

**L11.** 5.5.1 Importance of Assistive Technologies

**L12.** 5.5.2 Career Connection in Assistive Tech

# Application and Impacts of Computing

This chapter shows you how computing technologies such as Artificial Intelligence, the Internet of Things and Data Analytics are changing life in Pakistan — in cities, hospitals, farms and classrooms. You will also learn how to judge information sources, weigh the good and bad effects of computing, and see how assistive technologies give everyone an equal chance.

اس باب میں آپ سیکھیں گے کہ مصنوعی ذہانت (AI)، انٹرنیٹ آف تھنگز (IoT) اور ڈیٹا اینالیٹکس ہماری روزمرہ زندگی کو کیسے بدل رہے ہیں۔ آپ معلومات کے ذرائع کو پرکھنا، کمپیوٹنگ کے مثبت اور منفی اثرات سمجھنا، اور معذور افراد کی مدد کرنے والی ٹیکنالوجی کے بارے میں بھی جانیں گے۔

## What you will learn

- Explain what Artificial Intelligence (AI) is and give everyday examples of it.
- Name and explain the six components of an IoT system, with examples of each.
- Compare IoT, AI and Data Analytics by definition, purpose, type of work and output.
- Discuss the uses, benefits and challenges of AI in education in Pakistan.
- Analyse positive and negative impacts of computing in five fields, and practise responsible use.
- Describe how IoT, Data Analytics and AI work together in a smart system.
- List the five components of Data Analytics in the correct order.
- Describe uses of IoT in smart cities, industry, healthcare, education, homes and farming.
- Classify information sources as primary, secondary or tertiary, and judge when a source is unreliable.
- Explain the four categories of assistive technologies, their importance, and related careers.

## 5.1 Introduction to Computing Impacts

### LEARN IT

**Computing** means using computers and computer-based devices to store, process and share information. Today computing is an important part of your daily life. You use computers, mobile phones, the internet and smart devices almost everywhere — at home, in college and on the road.

In Pakistan, computing is becoming **vital** (very necessary) in:

- **Schools and colleges** — online admissions, e-books, online results of the Sindh boards.
- **Hospitals** — computerised patient records, digital X-rays, online appointment systems.
- **Banks** — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- **Farms** — weather forecasts on mobile phones, soil sensors, tube-well control.
- **Industries** — automatic machines in textile mills in Karachi, Hyderabad and Faisalabad.

An **impact** is the effect something has on people and society. Computing has **positive impacts** (speed, convenience, new jobs) and **negative impacts** (privacy risks, addiction, job loss). In this chapter you will study three key technologies first — AI, IoT and Data Analytics — and then their uses and impacts.

### WORKED EXAMPLE

**A day in Sara's life in Karachi — spot the computing:**

**Step 1.** 7:00 am — Her phone unlocks by recognising her face (AI).

**Step 2.** 7:30 am — She books a ride on a ride-hailing app; GPS shows the car's location (internet + location data).

**Step 3.** 9:00 am — College attendance is marked by a biometric (fingerprint) machine (IoT).

**Step 4.** 2:00 pm — She pays her fee through a mobile banking app (computing in business).

**Step 5.** 8:00 pm — YouTube suggests a physics lecture based on what she watched yesterday (AI).

Conclusion: in a single day, Sara uses computing in at least five different fields without even thinking about it.

## اردو میں سمجھیں

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔ پاکستان میں اسکول، اسپتال، بینک، کھیت اور صنعتیں سب کمپیوٹر اور انٹرنیٹ پر انحصار کرنے لگے ہیں۔ کمپیوٹنگ کے فائدے بھی ہیں اور نقصانات بھی، جنہیں ہم اس باب میں سمجھیں گے۔

### CHECK YOURSELF

1. Name four places in Pakistan where computing has become vital.
2. What is meant by the "impact" of computing?

Answers: 1. Any four of: schools, hospitals, banks, farms, industries. 2. The effect computing has on people and society — it can be positive or negative.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Computing is now part of:

- a. Daily life — phones, banks, schools
- b. Only museums
- c. Only NAND labs
- d. Only OSI textbooks

**MCQ 2.** An impact can be:

- a. Positive or negative
- b. Only positive
- c. Only hardware heat
- d. Only Big O

**MCQ 3.** This unit's three big technologies are:

- a. AI, IoT, Data analytics
- b. Only bubble sort
- c. Only Access forms
- d. Only Figma

**MCQ 4.** A smart meter is closest to:

- a. IoT
- b. A primary key
- c. A K-map
- d. A sprint

**MCQ 5.** Predicting fail risk from marks is:

- a. Data analytics / AI
- b. Physical cabling
- c. A pie only
- d. Waterfall always

**MCQ 6.** Students must also learn:

- a. Limits and ethics, not only hype
- b. That computers never fail
- c. That data is always true
- d. That IoT needs no internet

## Extra short questions

1. Give two daily computing uses in Pakistan.
2. Why study impacts, not only coding?
3. Name the three tools this chapter compares.
4. One negative impact.

## Extra long question

1. Write a balanced 10-mark intro: three benefits of computing in a Sindh college and three harms, with one mitigation each.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. JazzCash/Easypaisa; NADRA records; online boards; WhatsApp classrooms — any two.
2. Technology changes jobs, privacy, inequality — exams ask both sides.
3. AI, IoT, data analytics.
4. Job displacement, addiction, surveillance, e-waste.

**Long**

1. Benefits: attendance systems, online notes, analytics of results. Harms: distraction, data leaks, unfair access. Mitigations: phone policy, passwords/backups, lab hours for students without devices.

## 5.1.1 Artificial Intelligence (AI) ★ Golden

### LEARN IT

**Artificial Intelligence (AI)** is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

Understand the name in two parts:

- **Artificial** — it is *created by humans*; it is not natural.
- **Intelligence** — it can perform tasks that *require thinking*.

AI systems **use data to make decisions**. The more good data an AI system sees, the better it becomes at recognising patterns (repeating features) and predicting what to do.

**Examples from the source:**

1. **Face recognition** on a mobile phone to unlock the screen.
2. **YouTube suggesting videos** based on your interests.
3. **Chatbots** answering questions on a website (for example, a bank's help chat).

More Pakistani examples: Google Maps predicting traffic on Shahrah-e-Faisal, Urdu voice typing on your phone, banks flagging a suspicious card transaction, and AI tools that translate English into Urdu.

### AI in smart systems

AI rarely works alone. It often acts as the "**brain**" of a system, working together with the **Internet of Things (IoT)** and **Data Analytics**:

- **IoT devices gather data** — they are the "eyes" (sensors).
- **Analytics sorts the data** — it organises and studies it.
- **AI makes the final decision** — it is the "brain".
- An **action** follows — something happens in the real world.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

### WORKED EXAMPLE

#### Smart air-conditioner in a hot Hyderabad room — trace the system:

**Step 1 (IoT – eyes).** A temperature sensor reads the room temperature: 36 °C.

**Step 2 (Analytics).** The readings of the last 10 minutes are organised: 34, 35, 35, 36 °C — the room is getting hotter.

**Step 3 (AI – brain).** The AI compares this with the comfortable limit (say 26 °C) and with the fact that a person is in the room (motion sensor). Decision: "Turn the AC on."

**Step 4 (Action).** A switch turns the AC on automatically.

No human pressed a button — the system sensed, thought and acted.

### REMEMBER IT

**Eyes – Sort – Brain – Do.** IoT is the *eyes*, Analytics *sorts*, AI is the *brain*, and the actuator *does* the action. Think of a cricket fielder: his eyes see the ball (IoT), he judges speed and direction (analytics), decides to dive (AI), and his body moves (action).

### COMMON MISTAKE

**Mistake:** "AI means robots." **Fix:** Most AI is just *software* — YouTube suggestions, spam filters and face unlock have no robot body. A robot may *use* AI, but AI itself is the thinking software.

### اردو میں سمجھیں

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔ AI ڈیٹا کی مدد سے فیصلے کرتی ہے، جیسے چہرہ پہچان کر فون کھولنا یا یوٹیوب پر ویڈیوز تجویز کرنا۔ ایک اسمارٹ سسٹم میں IoT آنکھوں کا کام کرتا ہے اور AI دماغ کا۔

### EXAM TIP

Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks).

To score full marks: (1) write the two-part meaning of the name — *artificial* = *made by humans*, *intelligence* = *can think/decide*; (2) say "AI uses data to make decisions"; (3) give at least two examples; (4) for the "together" question, draw the flow IoT → Analytics → AI → Action and explain it with the room-AC example.

### CHECK YOURSELF

1. Why is AI called "artificial"?
2. In a smart system, which technology is the "brain" and which is the "eyes"?
3. Is a spam filter in Gmail an example of AI? Why?

---

Answers: 1. Because it is created by humans. 2. AI is the brain; IoT is the eyes. 3. Yes — it uses data (past spam emails) to decide whether a new email is spam, a task that needs "thinking".

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** AI is:

- a. Machines performing tasks that need human intelligence
- b. Any Excel sheet
- c. A copper cable
- d. A primary key

**MCQ 2.** Machine learning is AI that:

- a. Learns patterns from data
- b. Only follows a fixed 5-line script forever
- c. Is analog only
- d. Is a pie chart

**MCQ 3.** A chatbot is an:

- a. AI application
- b. OSI hub
- c. AND gate
- d. MVP canvas

**MCQ 4.** AI needs:

- a. Data + model + compute
- b. Only a mouse
- c. Only paper
- d. Only Agile stickers

**MCQ 5.** An exam risk of AI answers is:

- a. Hallucination / wrong confident text
- b. Too much NAND
- c. Too much OSI
- d. Too much bubble sort

**MCQ 6.** This is a ★ topic because papers ask:

- a. Definition + examples + one issue
- b. Only cabling colours
- c. Only Access wizards
- d. Only Big O proofs

## Extra short questions

1. Define AI with two examples.
2. AI vs ordinary software.
3. One benefit and one risk in exams/education.
4. Name two Pakistani/education uses.

## Extra long question

1. Explain AI to a Class XI student: definition, two components, two applications, two ethical issues. ★ Golden.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Intelligence in machines. Face unlock; Google Translate; board-result chatbots.
2. Ordinary: fixed rules. AI: improves from data / handles messy input.
3. Benefit: tutoring. Risk: cheating / wrong answers.
4. Urdu OCR; adaptive quizzes; NADRA biometrics (accept reasonable).

**Long**

1. Def. Data+algorithms(+sensors). Apps: health imaging, spam filter. Ethics: bias, privacy, job loss. One sentence on human remaining responsible.

## 5.1.2 Internet of Things (IoT) & Components ★ Golden

### LEARN IT

The **Internet of Things (IoT)** is a system in which **physical devices are connected to the Internet** and can **collect and share data** to perform tasks **automatically**.

The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so they can "talk" to each other and to us.

**Source example:** A **smartwatch** collects your heart-rate data and sends it to your mobile phone, where it is displayed.

### Components of IoT systems

IoT relies on **six foundational components** to work properly.

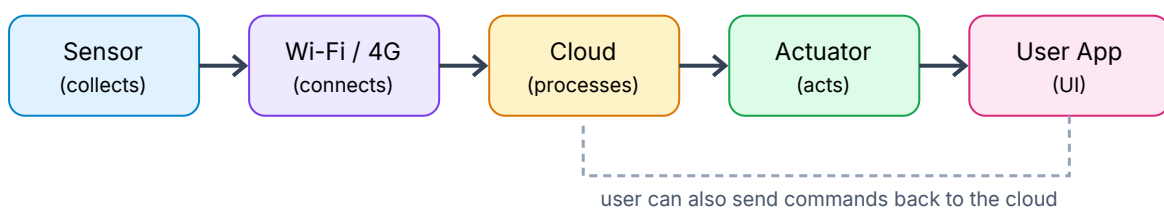
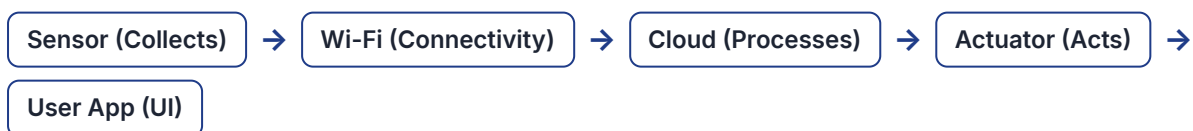
| # | Component              | What it does                                                                                                                      | Examples                                                                                  |
|---|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1 | <b>Sensors</b>         | Devices that <b>collect information</b> from the physical world. They are the "eyes and ears" of the system.                      | Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor |
| 2 | <b>Actuators</b>       | Devices that <b>perform physical actions</b> in response to system commands. They convert digital signals into physical movement. | Electric motor, switch, automatic door opener, water valve, fan controller                |
| 3 | <b>Connectivity</b>    | The connection that allows devices to <b>send and receive data</b> . Without it, sensors can't talk to the system.                | Wi-Fi, Bluetooth, ZigBee, 4G/5G                                                           |
| 4 | <b>Data Processing</b> | <b>Changing raw data into useful information.</b> Steps: collecting, organising, analysing and deciding.                          | Checking if the room temperature is too hot to trigger the AC                             |

|   |                            |                                                                                                                                |                                            |
|---|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| 5 | <b>Cloud Computing</b>     | <b>Storing data and using services over the internet</b> instead of local hard drives. Handles massive IoT data.               | Google Drive, iCloud, online LMS           |
| 6 | <b>User Interface (UI)</b> | The component that enables <b>interaction between the user and the device</b> . Helps users see information and give commands. | Mobile apps, touch screens, voice commands |

## Understanding each component

- **Sensors** turn something in the real world (heat, movement, a fingerprint, smoke, wetness of soil) into data. A smoke sensor in a Karachi shopping mall "smells" smoke and reports it.
- **Actuators** do the opposite: they take a digital command and make something *move* or *change*. The water valve that opens, the motor that turns, the door that slides open at a bank entrance.
- **Connectivity** is the "road" for data. Short distances use Bluetooth (watch to phone); a house uses Wi-Fi; low-power sensor networks use **ZigBee**; devices far away in fields use mobile networks (4G/5G from Jazz, Zong, Telenor/Ufone).
- **Data processing** is where raw numbers ("36") become meaning ("too hot — turn on AC"). Its four steps are *collect* → *organise* → *analyse* → *decide*.
- **Cloud computing** stores and processes the huge amount of data that thousands of sensors produce every second — far more than a single small device can hold.
- **User Interface** is how *you* see the data and give orders — an app on your phone, a touch screen on a smart fridge, or saying "Hey Google, turn off the lights".

## How the components connect



The six IoT components working as one system (data processing happens inside the cloud).

#### WORKED EXAMPLE

##### Smart fire alarm in a Karachi factory — identify all six components:

**Step 1 — Sensor:** A smoke sensor detects smoke near the machines.

**Step 2 — Connectivity:** It sends the reading over the factory Wi-Fi.

**Step 3 — Cloud computing:** The reading is stored on an online server along with the history of all sensors.

**Step 4 — Data processing:** The software checks: "Is the smoke level above the safe limit?" Yes → decide "Fire risk!"

**Step 5 — Actuator:** Water valves of the sprinklers open and the alarm bell switch turns on.

**Step 6 — User interface:** The manager's mobile app shows a red alert: "Smoke detected – Hall 3".

#### REMEMBER IT

**Sensor senses, Actuator acts.** Both words give you the answer: **sensor** = **see**/sense (input), **actuator** = **act** (output). Remember the six components with "**Some Ants Carry Data Carefully Up**" — Sensors, Actuators, Connectivity, Data processing, Cloud, UI.

#### COMMON MISTAKE

**Mistake:** Calling a fan controller or water valve a "sensor". **Fix:** Ask "does it *measure* something or *do* something?" A valve opens and closes — it *does*, so it is an **actuator**. A soil moisture sensor only *measures*, so it is a **sensor**.

#### اردو میں سمجھیں

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے ڈیٹا جمع اور شیئر کرتے ہیں۔ اس کے چھ بنیادی اجزاء ہیں: سینسرز (معلومات جمع کرتے ہیں)، ایکچویٹرز (جسمانی کام انجام دیتے ہیں جیسے موٹر چلانا یا دروازہ کھولنا)، کنیکٹیویٹی (وائی فائی، بلوٹوتھ)، ڈیٹا پروسیسنگ، کلاؤڈ کمپیوٹنگ، اور یوزر انٹرفیس (موبائل ایپ)۔

### EXAM TIP

"Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — *physical devices, connected to the internet, collect and share data automatically*. Then list all **six** components, each with a one-line meaning and one example. A labelled flow diagram (Sensor → Wi-Fi → Cloud → Actuator → App) earns extra credit.

Short questions often ask: "Differentiate between a sensor and an actuator" — sensor collects data (input, eyes and ears), actuator performs physical action (output, converts digital signals into movement).

### CHECK YOURSELF

1. Classify as sensor or actuator: (a) fingerprint scanner, (b) automatic door opener, (c) motion detector, (d) electric motor.
2. Name four types of connectivity used in IoT.
3. Why is cloud computing needed in IoT?

---

Answers: 1. (a) sensor, (b) actuator, (c) sensor, (d) actuator. 2. Wi-Fi, Bluetooth, ZigBee, 4G/5G. 3. Because IoT devices produce massive amounts of data; the cloud stores and processes it over the internet instead of on small local hard drives.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** IoT is:

- a. Physical devices on the internet sharing data
- b. Only desktop Word
- c. Only K-maps
- d. Only sprints

**MCQ 2.** A typical IoT chain is:

- a. Sensor → connectivity → data → action
- b. Query → form → report only
- c. SYN → ACK only
- d. Bubble → selection

**MCQ 3.** Connectivity might be:

- a. Wi-Fi, 4G, Bluetooth
- b. A primary key
- c. A pie chart
- d. A NAND-only bus

**MCQ 4.** An actuator:

- a. Does something in the real world (motor, relay)
- b. Only stores SQL
- c. Only sorts
- d. Only prints Urdu

**MCQ 5.** Without connectivity, sensors:

- a. Cannot send readings
- b. Become RDBMS
- c. Become AI models
- d. Become OSI layer 7

**MCQ 6.** A smart irrigation valve is IoT because:

- a. It senses, sends, and acts remotely
- b. It is made of paper
- c. It runs bubble sort
- d. It is a form wizard

## Extra short questions

1. Define IoT.
2. List four components.
3. Give a farm example.
4. One security worry.

## Extra long question

1. Draw the IoT component diagram for a smart classroom (lights, attendance, AC). Label each component and one risk. ★ Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Internet of Things: physical objects with sensors/software that connect and exchange data.
2. Devices/sensors, connectivity, data processing, user interface / actuators.
3. Soil moisture sensor → mobile network → app → pump on/off.
4. Unpatched cameras on the internet; weak default passwords.

### Long

1. Sensors (PIR, RFID), Wi-Fi, college server/analytics, dashboard, relays for lights. Risk: student location privacy. Mitigation: local processing, access control.

## 5.1.3 Data Analytics & Components

### LEARN IT

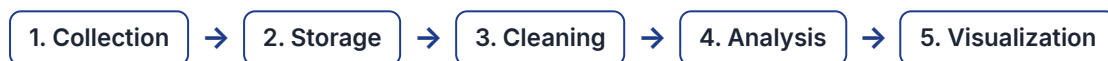
**Data Analytics** is the process of **collecting, organising, cleaning, studying and presenting data** to find useful information. It helps people make better decisions.

**Data** means raw facts and figures (for example, "72, 45, 88"). **Information** is data that has been processed so it has meaning ("Ali scored 72%; he passed"). Data Analytics is the bridge from data to information.

**Source example:** A college collects exam marks, calculates percentages, and identifies which students need extra help. This entire process is Data Analytics.

### Components of Data Analytics

Data Analytics comprises **five key components**, which happen in order like a pipeline:



1. **Data Collection** — Gathering raw information from multiple sources: websites, surveys, sensors, hospital records, bank transactions.
2. **Data Storage** — Storing data safely and properly after collection. It can be stored in files, databases, spreadsheets or the cloud.
3. **Data Cleaning** — Identifying and correcting errors in data. Raw data may contain:
  - **Duplication** of records (the same student entered twice),
  - **Incorrect values** (typos, e.g. marks "870" instead of "87"),
  - **Missing** files or records (a student's marks not entered),
  - **Improper formats** (dates written as 05/10/2026 in one place and "5 Oct 26" in another).
4. **Data Analysis** — Studying the clean data to identify patterns, trends and relevant information for problem-solving (e.g. finding out which subjects students failed the most).
5. **Data Visualization** — Creating visual representations of data (plots, charts, graphs) to provide valuable insights at a glance (e.g. a pie chart showing pass vs fail percentage).

### WORKED EXAMPLE

A college in Sukkur analyses Physics test results (out of 50) for 6 students.

**Step 1 — Collection:** Marks are gathered from the teacher's register.

| Roll No | Name   | Marks (raw) |
|---------|--------|-------------|
| 1       | Ayesha | 42          |
| 2       | Bilal  | 18          |
| 3       | Hina   | 35          |
| 3       | Hina   | 35          |
| 4       | Kashif | 250         |
| 5       | Sana   | (missing)   |

**Step 2 — Storage:** The marks are typed into a spreadsheet saved on Google Drive (cloud).

**Step 3 — Cleaning:** Hina's record is a *duplicate* → delete one copy. Kashif's 250 is an *incorrect value* (more than 50) → the register shows 25, so correct it. Sana's marks are *missing* → ask the teacher; she was absent, so mark her "absent".

**Step 4 — Analysis:** Clean marks: 42, 18, 35, 25. Pass mark is 17 (33% of 50 = 16.5, rounded up). All 4 passed. Percentages: Ayesha 84%, Bilal 36%, Hina 70%, Kashif 50%. Average =  $(42 + 18 + 35 + 25) \div 4 = 120 \div 4 = 30$  marks = 60%. Bilal is closest to failing → needs extra help.

**Step 5 — Visualization:** A bar chart shows each student's percentage; Bilal's short bar makes the problem visible at a glance.

### REMEMBER IT

**"Cows Sit Calmly And Vote"** — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar (store), pick out stones (clean), cook (analyse), serve beautifully on a plate (visualise).

### COMMON MISTAKE

**Mistake:** Skipping cleaning and going straight to analysis. **Fix:** Dirty data gives wrong answers. In the example, Kashif's wrong "250" would have made the average  $(42 + 18 + 35 + 250) \div 4 = 86.25$  — more than 50, which is impossible. Always clean first.

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔ اس کے پانچ اجزاء ہیں: ڈیٹا جمع کرنا، محفوظ کرنا، صاف کرنا (غلطیاں اور دہرائے گئے ریکارڈ ہٹانا)، تجزیہ کرنا، اور چارٹ یا گراف کی صورت میں دکھانا۔

#### CHECK YOURSELF

1. List the five components of Data Analytics in order.
2. Name four kinds of errors removed during data cleaning.
3. A pie chart of pass/fail percentage belongs to which component?

Answers: 1. Collection, Storage, Cleaning, Analysis, Visualization. 2. Duplicate records, incorrect values (typos), missing records, improper formats. 3. Data Visualization.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Data analytics is:

- a. Collect, clean, study, present data for decisions
- b. Only drawing NAND
- c. Only cabling
- d. Only Figma

**MCQ 2.** Cleaning data means:

- a. Fixing missing/wrong values
- b. Washing the PC
- c. Defrag only
- d. Sorting OSI layers

**MCQ 3.** A dashboard is:

- a. A visual summary of analytics
- b. A primary key
- c. A sprint log only
- d. A K-map

**MCQ 4.** Analytics without a question is:

- a. Just a pile of charts
- b. Always AI
- c. Always IoT
- d. Always SQL keys

**MCQ 5.** Finding that "second shift fails more in CS" is:

- a. Insight from analysis
- b. A foreign key
- c. A gate delay
- d. A beachhead

**MCQ 6.** Python's later tool for this in XII is:

- a. Pandas + charts
- b. LogiSim
- c. Figma only
- d. Access macros only

## Extra short questions

1. Define data analytics and list its steps.
2. Why clean first?
3. Give a college decision driven by analytics.
4. How is analytics different from IoT?

## Extra long question

1. A principal has 3 years of BIEK results. Outline an analytics project: source, clean, one chart, one decision.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

### **Short**

1. Collect → organise/clean → analyse → present → decide.
2. Garbage in, garbage out — mean of marks with blanks is wrong.
3. Extra classes for a subject whose fail rate jumped.
4. IoT gathers live device data; analytics is the studying/presenting (often of that data).

### **Long**

1. Source: gazette CSVs. Clean: missing roll numbers. Chart: line of pass% by year.  
Decision: change lab hours if practical marks drag theory. Mention ethics (no public shaming of named students).

## Table: IoT vs AI vs Data Analytics Comparison

### LEARN IT

These three technologies are often used together, so students mix them up. The table below, from the source, compares them on four aspects.

| Aspect                        | IoT                                                    | Artificial Intelligence (AI)                               | Data Analytics                                              |
|-------------------------------|--------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| <b>Definition</b>             | Network of physical devices connected to the internet. | Technology enabling machines to think and decide.          | Process of examining data to find patterns.                 |
| <b>Main purpose</b>           | Data collection and automation.                        | Intelligent decision making and predictions.               | Extracting meaningful insights.                             |
| <b>Type of work</b>           | Works with physical devices (hardware).                | Works with smart software systems.                         | Works with data (numbers, records).                         |
| <b>Output</b>                 | Raw data from the environment.                         | Smart decisions or actions.                                | Reports, graphs, insights.                                  |
| <b>Role in a smart system</b> | Eyes and hands                                         | Brain                                                      | Organiser / reporter                                        |
| <b>Pakistani example</b>      | Safe City cameras in Lahore/Karachi capturing traffic  | Software recognising a car's number plate for an e-challan | Monthly report showing which roads have the most violations |

### WORKED EXAMPLE

#### One problem, three technologies — a K-Electric smart meter:

**IoT:** The smart electricity meter measures units used every 15 minutes and sends them over the mobile network. Output: raw readings.

**Data Analytics:** The company cleans and studies a month of readings and draws a graph showing that most electricity is used between 7 pm and 11 pm. Output: report/graph.

**AI:** Software learns each area's pattern, predicts tomorrow's demand, and decides how to plan load management. Output: smart decision.

### REMEMBER IT

**Ask "What is the output?"** Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

#### COMMON MISTAKE

**Mistake:** Writing that IoT "makes decisions". **Fix:** In the comparison table IoT's main purpose is *data collection and automation*; the *intelligent decision* belongs to AI.

#### اردو میں سمجھیں

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔ AI ذہین سافٹ ویئر ہے جو فیصلے اور پیش گوئیاں کرتا ہے۔ ڈیٹا اینالیٹکس ڈیٹا کا جائزہ لے کر رپورٹس اور گراف بناتا ہے۔ تینوں مل کر ایک اسمارٹ نظام بناتے ہیں۔

#### CHECK YOURSELF

1. Which of the three works mainly with hardware?

2. What is the output of Data Analytics?

Answers: 1. IoT. 2. Reports, graphs and insights.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Data analytics is:

- a. Collect, clean, study, present data for decisions
- b. Only drawing NAND
- c. Only cabling
- d. Only Figma

**MCQ 2.** Cleaning data means:

- a. Fixing missing/wrong values
- b. Washing the PC
- c. Defrag only
- d. Sorting OSI layers

**MCQ 3.** A dashboard is:

- a. A visual summary of analytics
- b. A primary key
- c. A sprint log only
- d. A K-map

**MCQ 4.** Analytics without a question is:

- a. Just a pile of charts
- b. Always AI
- c. Always IoT
- d. Always SQL keys

**MCQ 5.** Finding that "second shift fails more in CS" is:

- a. Insight from analysis
- b. A foreign key
- c. A gate delay
- d. A beachhead

**MCQ 6.** Python's later tool for this in XII is:

- a. Pandas + charts
- b. LogiSim
- c. Figma only
- d. Access macros only

## Extra short questions

1. Define data analytics and list its steps.
2. Why clean first?
3. Give a college decision driven by analytics.
4. How is analytics different from IoT?

## Extra long question

1. A principal has 3 years of BIEK results. Outline an analytics project: source, clean, one chart, one decision.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

### **Short**

1. Collect → organise/clean → analyse → present → decide.
2. Garbage in, garbage out — mean of marks with blanks is wrong.
3. Extra classes for a subject whose fail rate jumped.
4. IoT gathers live device data; analytics is the studying/presenting (often of that data).

### **Long**

1. Source: gazette CSVs. Clean: missing roll numbers. Chart: line of pass% by year.  
Decision: change lab hours if practical marks drag theory. Mention ethics (no public shaming of named students).

## 5.2 Uses of IoT in Different Areas

### LEARN IT

IoT **improves efficiency** (doing more with less effort), **saves time**, and **enhances the overall quality of life**. It plays a significant role in developing modern **infrastructure** (the basic systems of a country — roads, power, water, hospitals). The source lists six areas.

### 1. Smart Cities

- **Traffic management:** Sensors adjust traffic signals automatically to reduce congestion (crowding). Example: signals on a busy Karachi road stay green longer for the direction with more cars.
- **Street lighting:** Lights turn ON/OFF based on daylight or movement, saving electricity.
- **Parking:** Sensors guide drivers to empty parking spaces, e.g. in a large shopping mall car park.
- **Waste management:** Smart bins send alerts when full, so trucks collect only full bins instead of driving to every bin.

### 2. Industrial Automation

- Sensors check machine **temperature and vibration** to prevent breakdowns (called predictive maintenance).
- Automated machines perform repeated tasks precisely.
- Quality-control sensors check products during manufacturing — for example, in the **textile industries of Sindh**, cameras and sensors spot faults in cloth as it is woven.

### 3. Healthcare (Smart Health)

- **Smartwatches** continuously measure heart rate and vital signs (pulse, oxygen level).
- **Remote monitoring devices** send patient data directly to doctors — vital for **rural areas of Pakistan**, where a specialist may be hundreds of kilometres away (e.g. a heart patient in Tharparkar monitored by a doctor in Karachi).
- **Emergency alert devices** inform hospitals instantly, for example when an elderly person falls.

### 4. Education (Smart Learning)

- **Smart classrooms:** digital boards and connected projectors.
- **Attendance systems:** biometric automated attendance for students and teachers (used in many Sindh government schools and colleges).
- **Resource management:** automated classroom lighting and AC control — off when the room is empty.

## 5. Home Automation

- **Smart lighting:** controlled remotely via mobile apps.
- **Security:** real-time monitoring via Wi-Fi cameras — you can watch your house gate on your phone.
- **Voice control:** using assistants (like Alexa or Google Assistant) to control TVs and appliances.

## 6. Smart Farming

Smart farming is **crucial for agriculture in Pakistan** (especially Sindh and Punjab) to **save water** and **improve yields** (the amount of crop produced).

- **Soil moisture sensors:** check if crops actually need water.
- **Automated irrigation:** turns water pumps ON/OFF based on soil data.
- **Drones:** monitor crop health from above, spotting disease or dry patches.

**Smart farming process:**



### WORKED EXAMPLE

**A cotton farmer near Nawabshah uses smart irrigation:**

**Step 1.** A soil moisture sensor in the field reads 18% moisture.

**Step 2.** It sends the reading over the 4G network to a cloud service.

**Step 3.** The cloud AI checks: cotton needs at least 25% moisture, and the weather forecast shows no rain → decision "Needs water: YES".

**Step 4.** The actuator (pump switch) turns the tube-well ON.

**Step 5.** When moisture reaches 30%, the AI decides "enough" and the pump turns OFF.

Result: water is used only when needed — important because Sindh faces water shortages — and the farmer sees everything on his phone app.

### REMEMBER IT

Remember the six areas with "**CHIEF H**": **C**ities, **H**ealth, **I**ndustry, **E**ducation, **F**arming, **H**omes.

### COMMON MISTAKE

**Mistake:** Giving only the name of the area ("smart cities") in an exam answer. **Fix:** Always add *what* IoT does there — e.g. "smart bins send alerts when full" — and, if possible, a Pakistani example.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔ اسمارٹ شہروں میں ٹریفک سگنل اور اسٹریٹ لائٹس خود کار ہوتی ہیں، اسپتالوں میں مریضوں کا ڈیٹا براہ راست ڈاکٹر تک پہنچتا ہے، اور کھیتوں میں مٹی کی نمی کے سینسر بتاتے ہیں کہ فصل کو پانی چاہیے یا نہیں، جس سے سندھ اور پنجاب میں پانی کی بچت ہوتی ہے۔

#### CHECK YOURSELF

1. How do smart bins help waste management?
2. Why is remote health monitoring especially useful in Pakistan?
3. In smart farming, which device is the actuator?

Answers: 1. They send alerts when full, so only full bins are emptied. 2. Rural areas lack doctors; devices send patient data directly to doctors far away. 3. The water pump (its switch).

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Smart cities use IoT for:

- a. Traffic, waste, lighting
- b. Only K-maps
- c. Only sprints
- d. Only Access reports

**MCQ 2.** Health IoT example:

- a. Remote heart-rate monitor
- b. Bubble sort
- c. Primary keys
- d. Figma frames

**MCQ 3.** Farms use IoT to:

- a. Water when soil is dry
- b. Compile C
- c. Draw ER
- d. Write OSI

**MCQ 4.** Homes: a smart lock is IoT if:

- a. It is on the network and reports/acts
- b. It is a mechanical padlock only
- c. It is a Python tuple
- d. It is a pie

**MCQ 5.** Education IoT:

- a. RFID attendance, smart boards
- b. Only textbooks
- c. Only NAND labs
- d. Only waterfalls

**MCQ 6.** A risk in all these areas:

- a. Privacy / hacked devices
- b. Too much paper
- c. Too few charts
- d. Too much Big O

## Extra short questions

1. Four application areas of IoT from the book.
2. One benefit in health.
3. One farm sensor + actuator pair.
4. Why security matters more as IoT grows.

## Extra long question

1. Pick city, hospital, farm. For each: one device, one data item, one action, one risk.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Cities, health, farms, homes, industry, education — any four.
2. Doctors see vitals without the patient travelling.

3. Moisture sensor + water pump.
4. More doors into private spaces (cameras, locks).

**Long**

1. City: traffic cam / count / signal / surveillance. Hospital: SpO2 / alert nurse / false alarm. Farm: moisture / pump / overwater if sensor fails.

## 5.2.1 Scope of AI in Education in Pakistan

### LEARN IT

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving **teaching**, **learning** and **administration** (managing the school: admissions, records, timetables).

#### Practical uses:

- **Learning Management Systems (LMS)** — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI features can recommend lessons to each student.
- **Automated grading and digital testing** — MCQ tests marked instantly by computer.
- **AI-based career counselling** — software that studies a student's marks and interests and suggests suitable fields.

#### Benefits:

- Helps students learn **at their own speed**.
- **Identifies students who need extra help**.
- Supports students with disabilities (**Assistive AI**), e.g. text-to-speech for a visually impaired student.

#### Major challenges in Pakistan:

- Limited **internet access in rural areas**.
- Lack of **computers and digital devices** for all students.
- Need for proper **teacher training** to use AI tools.
- **Data privacy and security** issues — students' personal data must be protected.

| Practical uses                      | Benefits                            | Challenges                |
|-------------------------------------|-------------------------------------|---------------------------|
| LMS                                 | Learn at own speed                  | Limited rural internet    |
| Automated grading & digital testing | Identifies students needing help    | Lack of computers/devices |
| AI career counselling               | Supports students with disabilities | Teacher training needed   |
| —                                   | —                                   | Data privacy & security   |

### WORKED EXAMPLE

#### Scenario — Fatima in Larkana and an AI tutor app:

**Step 1.** Fatima takes a 10-question online maths quiz on the college LMS. It is graded instantly (automated grading).

**Step 2.** The AI notices she got all 3 trigonometry questions wrong.

**Step 3.** It recommends two short videos and five easy practice questions on trigonometry — she learns at her own speed.

**Step 4.** Her teacher's dashboard flags "Fatima – needs help in trigonometry" (identifies students who need help).

**Challenge:** At home Fatima's village has weak 3G signals, so the videos buffer — showing the rural internet challenge.

### COMMON MISTAKE

**Mistake:** Writing that AI will replace teachers. **Fix:** The source says AI *improves* teaching and needs *teacher training* — teachers use AI as a tool; they are not replaced.

### اردو میں سمجھیں

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔ اس سے طلبہ اپنی رفتار سے سیکھ سکتے ہیں اور کمزور طلبہ کی نشاندہی ہوتی ہے۔ لیکن دیہی علاقوں میں انٹرنیٹ کی کمی، کمپیوٹرز کی قلت، اساتذہ کی تربیت اور ڈیٹا کی حفاظت بڑے چیلنج ہیں۔

### CHECK YOURSELF

1. Give three practical uses of AI in education.
2. State any two challenges of AI in education in Pakistan.

Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** AI in Pakistani classrooms can:

- a. Personalise practice, mark MCQs, translate
- b. Replace all teachers tomorrow by law
- c. Lay fibre
- d. Set primary keys

**MCQ 2.** A risk of AI homework tools is:

- a. Students submit work they cannot explain
- b. Faster bubble sort
- c. Better NAND
- d. Longer OSI

**MCQ 3.** Adaptive learning means:

- a. Difficulty changes with the student
- b. Everyone gets the same paper always
- c. No data is used
- d. Only analog TV

**MCQ 4.** Teachers still needed because:

- a. Care, judgment, fairness, lab skill
- b. AI cannot use electricity
- c. Books are illegal
- d. BIEK forbids computers

**MCQ 5.** Data used by education AI must be:

- a. Protected (student privacy)
- b. Posted publicly with names
- c. Sold always
- d. Ignored

**MCQ 6.** A realistic near-term use is:

- a. Urdu/English support and drill quizzes
- b. Fully autonomous colleges
- c. Replacing SDLC
- d. Replacing ER diagrams

## Extra short questions

1. Two uses of AI in education in Pakistan.
2. Two limits.
3. What should a student still be able to do unaided?
4. One policy idea for a college.

## Extra long question

1. Write a balanced note: opportunities of AI for BIEK CS students vs three rules a college should set.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Language practice; auto-marking MCQs; plagiarism hints; accessibility readers.
2. Bias, hallucinations, connectivity gaps, cheating.
3. Explain the answer in the board paper without the chatbot.
4. AI allowed for practice, banned in the exam hall; teach citation.

**Long**

1. Opp: extra drills, Urdu help, code explainers. Rules: no AI in exams, no pasting private data into public bots, teacher reviews flagged work. Link to digital literacy.

## 5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

### LEARN IT

**Information sources** are ways to find knowledge and data. The internet has vast information, but **not all of it is accurate**. Sources are divided into three main categories.

**1. Primary sources** give **original, firsthand information** straight from people, events or official records. They are **highly authentic** (genuine).

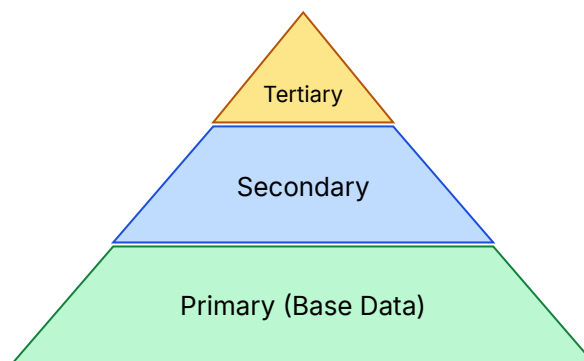
Examples: interviews, photographs, diaries, NADRA official records.

**2. Secondary sources** contain information that has **already been collected and explained by someone else**. They **summarise or analyse primary data**.

Examples: textbooks, news reports, research papers, Pakistan Bureau of Statistics reports.

**3. Tertiary sources** provide **highly summarised and organised information** derived from both primary and secondary sources. They are used for **quick reference**.

Examples: encyclopedias, Wikipedia, dictionaries, handbooks.



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

### Many examples, sorted

| Primary (firsthand)                   | Secondary (explained by others)                            | Tertiary (summarised, quick reference)  |
|---------------------------------------|------------------------------------------------------------|-----------------------------------------|
| Interview with a flood victim in Dadu | Newspaper report on the 2022 floods                        | Encyclopedia entry "Floods in Pakistan" |
| Photograph taken at an event          | Textbook chapter explaining the event                      | Wikipedia article                       |
| Personal diary or letter              | Biography written by a historian                           | Dictionary                              |
| NADRA official records (CNIC data)    | Pakistan Bureau of Statistics report analysing census data | Handbook / almanac of facts             |

|                                             |                                     |                                  |
|---------------------------------------------|-------------------------------------|----------------------------------|
| Original survey responses                   | Research paper analysing the survey | Index or list of research papers |
| Sensor readings from an IoT weather station | TV news analysis of the weather     | A school "fact sheet" on climate |
| Quaid-e-Azam's original speech recording    | Article discussing the speech       | Encyclopedia summary of his life |

## When do sources become unreliable?

In the digital age you must check facts carefully. A source is **unreliable** (cannot be trusted) if:

| Source type      | Reasons for unreliability                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------|
| <b>Primary</b>   | Information is incomplete, data is recorded incorrectly, or false information was deliberately provided. |
| <b>Secondary</b> | The information is outdated, lacks proper references/citations, or has not been verified by an expert.   |
| <b>Tertiary</b>  | Comes from an unknown/unverified website, or the information is oversimplified and misses key facts.     |

### REMEMBER IT

**Rule of thumb (Author – Date – References):** If an online article does not state the **author's name**, the **date of publication**, or list its **references**, it should not be trusted for academic work. Remember "**ADR**" — like checking a bus has a driver, a route and a ticket before you board.

### WORKED EXAMPLE

**Classify these — worked solutions:**

**(a) Your grandfather's handwritten diary from 1947.** Written by a person who lived through the event → firsthand → **Primary**.

**(b) A Dawn newspaper article analysing the budget.** A journalist collected facts and explained them → **Secondary**.

**(c) An Oxford English–Urdu dictionary.** Organised, summarised reference built from many sources → **Tertiary**.

**(d) A Bureau of Statistics report on literacy in Sindh.** It analyses census (primary) data → **Secondary**.

**(e) A photo you took of a road accident.** Original record of the event → **Primary**.

Method: Ask (1) "Was the creator *there* / is it the original record?" → Primary. (2) "Is someone *explaining or analysing* original data?" → Secondary. (3) "Is it a *quick-reference summary* of many sources?" → Tertiary.

### CHECK YOURSELF

**Exercise — classify these as Primary (P), Secondary (S) or Tertiary (T):**

1. A video interview with a cricketer after a match.
2. Your Computer Science textbook.
3. Wikipedia page on "Mohenjo-daro".
4. A birth certificate issued by the union council.
5. A research paper reviewing ten studies on smog in Lahore.
6. A handbook of physics formulas.
7. Original exam answer sheets of students.
8. A TV news report summarising a court decision.

---

Answers: 1. P 2. S 3. T 4. P 5. S 6. T 7. P 8. S

### COMMON MISTAKE

**Mistake:** Thinking "primary" means "most important" or "tertiary" means "useless". **Fix:** The names describe *how far the information is from the original event*. Primary = straight from the source; tertiary = summarised summary. Each is useful for a different job. Also, Wikipedia is **tertiary** — not primary — because it summarises other sources.

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔ پرائمری ذرائع براہ راست اور اصل معلومات دیتے ہیں (جیسے انٹرویو، تصاویر، نادرا کا ریکارڈ)۔ سیکنڈری ذرائع وہ ہیں جن میں کسی اور نے پرائمری معلومات کی وضاحت یا تجزیہ کیا ہو (جیسے درسی کتابیں، خبریں)۔ ٹرشییری ذرائع خلاصہ شدہ معلومات دیتے ہیں جیسے انسائیکلوپیڈیا، وکی پیڈیا اور لغت۔ جس آن لائن مضمون میں مصنف کا نام، تاریخ یا حوالہ جات نہ ہوں اس پر بھروسہ نہ کریں۔

#### EXAM TIP

Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short), and "When does a source become unreliable?"

Write the definition key phrase for each — *firsthand/original, collected and explained by someone else, highly summarised for quick reference* — plus two examples each. Draw the pyramid with primary at the base. For unreliability, give one reason for each type and quote the author–date–references rule.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** A primary source is:

- a. Original, first-hand
- b. A summary of summaries
- c. Always Wikipedia
- d. A pie chart of rumours

**MCQ 2.** A secondary source is:

- a. Someone else's analysis of primary material
- b. Raw sensor logs only
- c. Always false
- d. A NAND gate

**MCQ 3.** A tertiary source is:

- a. Highly summarised (encyclopaedia, textbook index)
- b. A lab notebook
- c. A witness interview
- d. A packet capture

**MCQ 4.** Your own class survey is:

- a. Primary
- b. Tertiary
- c. Always secondary
- d. Not a source

**MCQ 5.** A newspaper report of that survey is:

- a. Secondary
- b. Primary always
- c. Tertiary always
- d. IoT

**MCQ 6.** Board papers want you to:

- a. Judge reliability, not just copy Google
- b. Use any blog
- c. Ignore dates
- d. Prefer unsigned forwards

## Extra short questions

1. Define the three source types with one example each.
2. Which is a BIEK past paper for you as a student?
3. One reliability test.
4. Why tertiary is still useful.

## Extra long question

1. For a project "screen time vs grades", list two primary, two secondary, one tertiary source and how you would check each. ★ Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Primary: experiment, interview, sensor. Secondary: review article, textbook chapter discussing others. Tertiary: encyclopaedia, library catalogue.
2. Primary evidence of the exam; or secondary if you only read a guide about it — say so.
3. Author, date, purpose, evidence, bias.
4. Fast overview and pointers to better sources — not the final citation for a claim.

### Long

1. Primary: your survey, school lab logs. Secondary: a journalistic feature, a review paper. Tertiary: textbook definition of survey. Checks: sample size, date, conflict of interest.

## 5.4 Impacts of Computing in Various Fields

### LEARN IT

Computing has drastically changed society. It makes life **faster and more connected**, but it also brings challenges like **privacy risks** and **technology addiction**. For every field, think of both sides.

| Field         | Positive impact (+)                                                                                                                 | Negative impact (–)                                                                                      |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Education     | Online classes, e-books and learning apps make studying highly flexible and accessible.                                             | Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain). |
| Communication | Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai). | People interact less in person, leading to social isolation and reduced face-to-face skills.             |
| Business      | Online shopping, digital banking and automation dramatically improve speed and efficiency.                                          | Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.                     |
| Healthcare    | Digital patient records and telemedicine allow doctors to treat remote patients accurately.                                         | Patient data privacy is at risk if hospital servers are hacked.                                          |
| Agriculture   | Smart irrigation and weather forecasting massively increase crop yields.                                                            | Poor internet access in rural areas means many farmers cannot use these tools.                           |

### Smart and responsible use

- **Protect personal information** and use **strong passwords** (long, with letters, numbers and symbols; never share your OTP or CNIC details with callers).
- **Do not share fake news** or unverified WhatsApp forwards.
- **Balance screen time** with physical sports and activities.

### WORKED EXAMPLE

#### Scenario analysis — a bank in Karachi moves to mobile banking:

**Step 1 — Identify the field:** Business (banking).

**Step 2 — Positive:** Customers transfer money in seconds from home; no queue at the branch; the bank processes more transactions with fewer staff (speed and efficiency).

**Step 3 — Negative:** Some cashier jobs are no longer needed (job loss); hackers may attempt cyber-attacks or phishing SMS to steal accounts.

**Step 4 — Responsible use:** Customers should use strong passwords, never share OTPs, and report suspicious messages.

### REMEMBER IT

For the five fields, remember "E-C-B-H-A" — Education, Communication, Business, Healthcare, Agriculture — and always write one + and one – for each.

### COMMON MISTAKE

**Mistake:** Writing only positive impacts in a "discuss the impacts" question. **Fix:** "Impact" means *both* sides. Examiners expect a balanced answer: advantages, disadvantages, and how to use technology responsibly.

### اردو میں سمجھیں

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔ تعلیم، مواصلات، کاروبار، صحت اور زراعت میں اس نے کام تیز اور آسان بنائے ہیں، لیکن اس کے ساتھ اسکرین کی لت، سماجی تنہائی، ملازمتوں کا خاتمہ، بیکنگ اور پرائیویسی کے خطرات بھی پیدا ہوئے ہیں۔ ذمہ دارانہ استعمال کے لیے مضبوط پاس ورڈ رکھیں، جعلی خبریں نہ پھیلائیں، اور کھیل کود کے ساتھ اسکرین ٹائم میں توازن رکھیں۔

### CHECK YOURSELF

1. Give one positive and one negative impact of computing in healthcare.
2. Why can't many farmers in Pakistan benefit from smart agriculture tools?
3. State three rules of smart and responsible use.

Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked. 2. Poor internet access in rural areas. 3. Protect personal information/use strong passwords; do not share fake news; balance screen time with physical activity.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Computing in health:

- a. Records, imaging, telemedicine
- b. Only games
- c. Only NAND
- d. Only sprints

**MCQ 2.** Computing in education:

- a. LMS, simulations, analytics
- b. Only chalk forever
- c. Only OSI layer 1
- d. Only keys

**MCQ 3.** A social harm is:

- a. Addiction / fake news / job loss
- b. Faster search in a library
- c. Safer banks always only
- d. Shorter SDLC only

**MCQ 4.** E-waste is:

- a. Discarded electronics
- b. A Python error
- c. A foreign key
- d. A histogram bin

**MCQ 5.** The exam pattern is:

- a. One field + one good + one bad + one control
- b. Only definitions
- c. Only code
- d. Only careers

**MCQ 6.** Banks use computing for:

- a. Accounts, ATMs, fraud analytics
- b. Growing wheat
- c. Drawing K-maps
- d. Sorting hats

## Extra short questions

1. Two fields and one impact each.
2. One environmental issue.
3. One social issue.
4. One control/mitigation.

## Extra long question

1. "Computing has only benefits." Refute with three fields, each with a benefit, a harm, and a mitigation. 10 marks.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Health: telemedicine. Transport: ride apps / signals. Banking: mobile wallets.
2. E-waste, energy of data centres.
3. Digital divide — students without smartphones.
4. Laws, privacy settings, recycling, lab access at college.

**Long**

1. Health, education, media. Structure a table. Close: technology is a tool; policy and literacy decide the outcome.

## 5.5 Assistive Technologies ★ Golden

### LEARN IT

**Assistive technologies** are **devices, software and systems** that help people with **disabilities or functional limitations** (difficulty in seeing, hearing, moving, speaking, remembering, etc.). They allow more people to **participate equally** in school, work and society.

There are **four categories**:

| Category                            | Purpose                                                            | Examples                                                   |
|-------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|
| 1. <b>Mobility aids</b>             | Help people move from one place to another.                        | Smart wheelchairs, stair lifts                             |
| 2. <b>Sensory aids</b>              | Support vision and hearing to improve interaction.                 | Hearing aids, screen readers (VoiceOver), Braille displays |
| 3. <b>Communication aids</b>        | Assist individuals with speech difficulties to express themselves. | Speech-generating devices, text-to-speech apps             |
| 4. <b>Cognitive / learning aids</b> | Help people with memory, reading and understanding.                | Audiobooks, spell-checkers, digital reminder apps          |

Mobility · Sensory · Communication · Cognitive

- A **screen reader** is software that reads aloud the text on a screen, so a blind person can use a phone or computer. **VoiceOver** is Apple's screen reader; Android has TalkBack.
- A **Braille display** is a device with small pins that rise and fall to form Braille dots, so a blind user can read screen text with their fingers.
- A **speech-generating device** speaks out words that a person types or selects on a screen.
- A **smart wheelchair** can be controlled by a joystick, voice or head movement and may avoid obstacles with sensors.

### WORKED EXAMPLE

#### Scenario — choose the right assistive technology:

(a) Ahmed, a student in Hyderabad, is blind and wants to read his e-textbook. → **Sensory aid**: a screen reader or Braille display.

(b) Mrs Rashida cannot climb the stairs of her house after a knee operation. → **Mobility aid**: stair lift (or smart wheelchair).

(c) Zain cannot speak clearly due to a medical condition but can type. → **Communication aid**: text-to-speech app or speech-generating device.

(d) Nani often forgets to take her medicine. → **Cognitive aid**: digital reminder app.

(e) Usman has dyslexia (difficulty reading). → **Cognitive/learning aid**: audiobooks and spell-checkers.

Method: identify the *limitation* (moving, seeing/hearing, speaking, memory/reading) → choose the matching category.

### REMEMBER IT

Four categories = "**My Small Cat Cries**": **Mobility**, **Sensory**, **Communication**, **Cognitive**.  
Link each to a body function: *legs* (mobility), *eyes/ears* (sensory), *mouth* (communication), *mind* (cognitive).

### COMMON MISTAKE

**Mistake**: Putting text-to-speech and screen readers in the same category every time.

**Fix**: In the source, the **screen reader** is a *sensory* aid (supports vision), while a **text-to-speech app** is listed as a *communication* aid (helps a person with speech difficulty express themselves). Decide by *whose problem* it solves.

### اردو میں سمجھیں

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور معاشرے میں برابر حصہ لے سکیں۔ ان کی چار اقسام ہیں: نقل و حرکت میں مدد (اسمارٹ وہیل چیئر)، حسی مدد (سماعتی آلہ، اسکرین ریڈر، بریل ڈسپلے)، گفتگو میں مدد (اسپیج جنریٹنگ ڈیوائس) اور ذہنی و تعلیمی مدد (آڈیو بکس، اسپیل چیکر، یاد دہانی ایپس)۔

#### EXAM TIP

Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of \_\_\_\_" (MCQ).

For full marks write the definition (devices, software and systems; help people with disabilities; equal participation), then all **four** categories each with purpose + two examples. Finish with one line on importance (equal access, independence, inclusion).

#### CHECK YOURSELF

1. Which category does a hearing aid belong to?
2. Give two examples of cognitive/learning aids.
3. Define assistive technologies in one sentence.

---

Answers: 1. Sensory aids. 2. Any two: audiobooks, spell-checkers, digital reminder apps. 3. Devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Assistive technology helps:

- a. People with disabilities use systems
- b. Only hackers
- c. Only DBMS keys
- d. Only Agile coaches

**MCQ 2.** A screen reader is for:

- a. Blind / low-vision users
- b. Faster bubble sort
- c. Wi-Fi
- d. Primary keys

**MCQ 3.** Captions help:

- a. Deaf / hard-of-hearing users (and many others)
- b. Routers
- c. NAND gates
- d. Pie charts only

**MCQ 4.** Importance: they support:

- a. Equal access to education and work
- b. Faster CPUs only
- c. Cheaper RAM only
- d. OSI layer 1 only

**MCQ 5.** A career path is:

- a. Accessibility specialist / AT developer
- b. Only analog radio
- c. Only waterfalls
- d. Only K-maps

**MCQ 6.** Good HCI + AT means:

- a. Design for a wide range of users from the start
- b. A separate broken version forever
- c. No keyboard needed ever
- d. No contrast ever

## Extra short questions

1. Define assistive technology with two examples.
2. Why it matters in a Pakistani college.
3. Link to HCI accessibility principles.
4. One career connection from the book.

## Extra long question

1. Propose AT + ordinary design changes so a student with low vision and a student with limited hands can use the college portal. 10 marks. ★ if tagged Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

**Short**

1. Devices/software that help functional limits: screen readers, switches, captions, magnification.
2. A blind student can still take CS if materials are readable by NVDA and papers have electronic copies.
3. Perceivable, operable, understandable, robust — AT relies on those.
4. AT designer, accessibility tester, special-education technologist.

**Long**

1. Low vision: contrast, zoom, screen reader labels. Limited hands: large targets, keyboard/switch, extra time. Test with real users. Do not ship a mouse-only form.

## 5.5.1 Importance of Assistive Technologies

### LEARN IT

Assistive technologies are important for three main reasons:

1. **Equal access to education:** Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
2. **Independence in daily life:** Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the market without someone pushing.
3. **Social inclusion:** They break down barriers, allowing individuals to work, study and socialise confidently.

Equal Access → Independence → Inclusion

### WORKED EXAMPLE

**How one tool changes a life:** A visually impaired student in Karachi uses a screen reader on her laptop. **Equal access** — she reads the same e-books as her class. **Independence** — she no longer needs someone to read to her. **Inclusion** — she joins the class WhatsApp group, takes online tests and later gets a job, contributing to society.

### REMEMBER IT

Remember **"All"**: Access, Independence, Inclusion.

### اردو میں سمجھیں

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی میں خود مختاری (کسی کی مسلسل مدد کی ضرورت کم ہوتی ہے)، اور سماجی شمولیت (لوگ اعتماد کے ساتھ پڑھ، کام اور میل جول کر سکتے ہیں)۔

### CHECK YOURSELF

1. State three reasons why assistive technologies are important.
2. How do mobility aids support independence?

Answers: 1. Equal access to education, independence in daily life, social inclusion. 2. They reduce the need for constant human assistance.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Assistive technology helps:

- a. People with disabilities use systems
- b. Only hackers
- c. Only DBMS keys
- d. Only Agile coaches

**MCQ 2.** A screen reader is for:

- a. Blind / low-vision users
- b. Faster bubble sort
- c. Wi-Fi
- d. Primary keys

**MCQ 3.** Captions help:

- a. Deaf / hard-of-hearing users (and many others)
- b. Routers
- c. NAND gates
- d. Pie charts only

**MCQ 4.** Importance: they support:

- a. Equal access to education and work
- b. Faster CPUs only
- c. Cheaper RAM only
- d. OSI layer 1 only

**MCQ 5.** A career path is:

- a. Accessibility specialist / AT developer
- b. Only analog radio
- c. Only waterfalls
- d. Only K-maps

**MCQ 6.** Good HCI + AT means:

- a. Design for a wide range of users from the start
- b. A separate broken version forever
- c. No keyboard needed ever
- d. No contrast ever

## Extra short questions

1. Define assistive technology with two examples.
2. Why it matters in a Pakistani college.
3. Link to HCI accessibility principles.
4. One career connection from the book.

## Extra long question

1. Propose AT + ordinary design changes so a student with low vision and a student with limited hands can use the college portal. 10 marks. ★ if tagged Golden.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Devices/software that help functional limits: screen readers, switches, captions, magnification.
2. A blind student can still take CS if materials are readable by NVDA and papers have electronic copies.
3. Perceivable, operable, understandable, robust — AT relies on those.
4. AT designer, accessibility tester, special-education technologist.

**Long**

1. Low vision: contrast, zoom, screen reader labels. Limited hands: large targets, keyboard/switch, extra time. Test with real users. Do not ship a mouse-only form.

## 5.5.2 Career Connection in Assistive Tech

### LEARN IT

Designing assistive technology is a **highly rewarding career path** — you earn a living while improving people's lives.

| Career                              | What they do                                                                                                                                 |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assistive Software Developer</b> | Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).                                 |
| <b>Biomedical Engineer</b>          | Designs artificial limbs and hearing aids.                                                                                                   |
| <b>UX/UI Accessibility Expert</b>   | Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them). |

**UX** means *user experience* (how easy and pleasant a product is to use); **UI** means *user interface* (the screens and buttons you interact with). **Accessibility** means designing so that *everyone*, including people with disabilities, can use it.

### اردو میں سمجھیں

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔ اسسٹو سافٹ ویئر ڈویلپر اسپیشل ٹو ٹیکسٹ جیسے ٹولز بناتا ہے، بائیو میڈیکل انجینئر مصنوعی اعضاء اور سماعتی آلات تیار کرتا ہے، اور UX/UI ایکسیسبیلٹی ماہر یقینی بناتا ہے کہ ویب سائٹس نابینا افراد کے لیے بھی آسانی سے استعمال ہو سکیں۔

### CHECK YOURSELF

1. Who designs artificial limbs and hearing aids?
2. What does a UX/UI accessibility expert ensure?

Answers: 1. A biomedical engineer. 2. That websites are easy to navigate for visually impaired users.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Assistive technology helps:

- a. People with disabilities use systems
- b. Only hackers
- c. Only DBMS keys
- d. Only Agile coaches

**MCQ 2.** A screen reader is for:

- a. Blind / low-vision users
- b. Faster bubble sort
- c. Wi-Fi
- d. Primary keys

**MCQ 3.** Captions help:

- a. Deaf / hard-of-hearing users (and many others)
- b. Routers
- c. NAND gates
- d. Pie charts only

**MCQ 4.** Importance: they support:

- a. Equal access to education and work
- b. Faster CPUs only
- c. Cheaper RAM only
- d. OSI layer 1 only

**MCQ 5.** A career path is:

- a. Accessibility specialist / AT developer
- b. Only analog radio
- c. Only waterfalls
- d. Only K-maps

**MCQ 6.** Good HCI + AT means:

- a. Design for a wide range of users from the start
- b. A separate broken version forever
- c. No keyboard needed ever
- d. No contrast ever

## Extra short questions

1. Define assistive technology with two examples.
2. Why it matters in a Pakistani college.
3. Link to HCI accessibility principles.
4. One career connection from the book.

## Extra long question

1. Propose AT + ordinary design changes so a student with low vision and a student with limited hands can use the college portal. 10 marks. ★ if tagged Golden.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Devices/software that help functional limits: screen readers, switches, captions, magnification.
2. A blind student can still take CS if materials are readable by NVDA and papers have electronic copies.
3. Perceivable, operable, understandable, robust — AT relies on those.
4. AT designer, accessibility tester, special-education technologist.

**Long**

1. Low vision: contrast, zoom, screen reader labels. Limited hands: large targets, keyboard/switch, extra time. Test with real users. Do not ship a mouse-only form.

# Application and Impacts of Computing

12 single-page lessons • 4 Golden • read, close, say aloud, then sit the exam

## Chapter in one look

- Computing is vital in Pakistan's schools, hospitals, banks, farms and industries; it has both positive and negative impacts.
- **AI** is "artificial" (made by humans) and "intelligent" (does tasks needing thinking); it uses data to learn, recognise patterns and make decisions.
- In smart systems, IoT is the "eyes" that gather data, Analytics sorts it, and AI is the "brain" that decides the action.
- **IoT** is a system of physical devices connected to the internet that collect and share data to perform tasks automatically.
- The six IoT components are sensors (collect), actuators (act), connectivity, data processing, cloud computing and user interface.
- **Data Analytics** is a pipeline: collection → storage → cleaning → analysis → visualization.
- Data cleaning removes duplicates, incorrect values, missing records and improper formats.
- IoT outputs raw data (hardware); AI outputs smart decisions (software); Data Analytics outputs reports, graphs and insights (data).
- IoT is used in smart cities, industrial automation, healthcare, education, home automation and smart farming.
- Smart farming (soil sensor → cloud AI → water pump) saves water and improves yields in Sindh and Punjab.
- AI in Pakistani education: LMS, automated grading, career counselling; challenges are rural internet, lack of devices, teacher training and data privacy.
- IoT/AI careers (IoT engineer, data analyst) need programming, networking, electronics and problem-solving.
- Information sources: primary (firsthand/original), secondary (collected and explained by others), tertiary (summarised for quick reference).
- Do not trust an online article that lacks an author, a publication date or references.
- Computing helps education, communication, business, healthcare and agriculture but risks screen addiction, isolation, job loss, cyber-attacks and privacy loss.
- **Assistive technologies** — mobility, sensory, communication and cognitive aids — give equal access, independence and social inclusion to people with disabilities.

## Key terms

| Term                         | Meaning                                                                                       | اردو                               |
|------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------|
| Computing                    | Using computers and digital devices to store, process and share information.                  | کمپیوٹنگ                           |
| Artificial Intelligence (AI) | Human-made technology that lets machines think, learn and decide using data.                  | مصنوعی ذہانت                       |
| Internet of Things (IoT)     | Physical devices connected to the internet that collect and share data to work automatically. | انٹرنیٹ آف تھنگز، اشیاء کا انٹرنیٹ |
| Sensor                       | Device that collects information from the physical world.                                     | حساس آلہ (سینسر)                   |

|                     |                                                                                          |                        |
|---------------------|------------------------------------------------------------------------------------------|------------------------|
| Actuator            | Device that performs a physical action in response to a command.                         | عمل کار آلہ (ایکچویٹر) |
| Connectivity        | The link (Wi-Fi, Bluetooth, ZigBee, 4G/5G) that lets devices send and receive data.      | رابطہ کاری             |
| Data processing     | Changing raw data into useful information.                                               | ڈیٹا کی پروسیسنگ       |
| Cloud computing     | Storing data and using services over the internet instead of local drives.               | کلاؤڈ کمپیوٹنگ         |
| User interface (UI) | The part that lets a user see information and give commands to a device.                 | صارف انٹرفیس           |
| Data Analytics      | Collecting, storing, cleaning, analysing and presenting data to find useful information. | ڈیٹا کا تجزیہ          |
| Data cleaning       | Finding and correcting errors such as duplicates and missing values.                     | ڈیٹا کی صفائی          |
| Data visualization  | Showing data as charts, plots or graphs.                                                 | ڈیٹا کی بصری پیشکش     |

#### CHAPTER 5 EXAM · CLOSED BOOK

## Application and Impacts of Computing

75 marks · 1 hour 15 minutes · no notes · no phones

#### CHAPTER EXAM RULES

**Instructions.** Sit this exam after Lesson 12 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

**Marks.** 15 + 30 + 30 = 75. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

### Section A — Multiple choice (15 × 1 = 15)

1. AI is called "artificial" because it:

- a. runs only on electricity
- b. is created by humans
- c. is always wrong
- d. works without data

2. Which of the following is an actuator?
  - a. Temperature sensor
  - b. Motion sensor
  - c. Water valve
  - d. Wi-Fi router
3. ZigBee is an example of which IoT component?
  - a. Connectivity
  - b. Sensor
  - c. User interface
  - d. Actuator
4. Google Drive and iCloud are examples of:
  - a. Sensors
  - b. Data cleaning
  - c. Actuators
  - d. Cloud computing
5. Removing duplicate student records belongs to which component of Data Analytics?
  - a. Data collection
  - b. Data storage
  - c. Data cleaning
  - d. Data visualization
6. A pie chart showing pass vs fail percentage is an example of:
  - a. Data collection
  - b. Data storage
  - c. Data cleaning
  - d. Data visualization
7. According to the comparison table, the output of IoT is:
  - a. Raw data from the environment
  - b. Smart decisions
  - c. Reports and graphs
  - d. Printed documents
8. In a smart system, AI acts as the:
  - a. eyes
  - b. brain
  - c. hands
  - d. wires
9. NADRA's official records are a:
  - a. tertiary source
  - b. secondary source
  - c. primary source
  - d. unreliable source
10. Wikipedia is classified as a:
  - a. primary source
  - b. tertiary source
  - c. secondary source
  - d. firsthand record
11. Smart bins that send alerts when full are a use of IoT in:
  - a. smart farming
  - b. healthcare
  - c. home automation
  - d. smart cities
12. A system in which a soil sensor tells the cloud AI to switch on a water pump is used in:
  - a. smart farming
  - b. industrial automation
  - c. smart classrooms
  - d. traffic management

3. A screen reader such as VoiceOver is a:
- a. mobility aid
  - b. communication aid
  - c. sensory aid
  - d. cognitive aid
4. A speech-generating device is a:
- a. mobility aid
  - b. cognitive aid
  - c. sensory aid
  - d. communication aid
5. An online article should NOT be trusted for academic work if it lacks:
- a. colourful pictures
  - b. the author's name, publication date or references
  - c. a long title
  - d. advertisements

## Section B — Short questions (10 × 3 = 30)

---

1. Define Artificial Intelligence and give two examples from daily life.
2. Define the Internet of Things (IoT) with one example.
3. Differentiate between a sensor and an actuator with two examples of each.
4. What is the role of cloud computing in an IoT system?
5. List the five components of Data Analytics. What kinds of errors are removed in data cleaning?
6. Give two uses of IoT in smart cities and two in healthcare.
7. State two benefits and two challenges of AI in education in Pakistan.
8. Differentiate between primary and secondary sources with two examples each.
9. When does a secondary source become unreliable?
10. Give one positive and one negative impact of computing on (a) communication and (b) business.

## Section C — Long questions (attempt any 3 × 10 = 30)

---

1. Define IoT. Explain its six components with an example of each. Draw a diagram showing how they connect.
2. Compare IoT, AI and Data Analytics in a table (definition, purpose, type of work, output). Then explain, with an example, how the three work together in a smart system.
3. A college in Mirpurkhas wants to find out why many students are failing Mathematics. Explain how it would use each of the five components of Data Analytics to solve this problem.
4. Explain the three types of information sources with examples. Classify the following: (i) a school textbook, (ii) a photograph of an event, (iii) an encyclopedia, (iv) an interview, (v) a Bureau of Statistics report, (vi) a dictionary. When does each type become unreliable?

5. What are assistive technologies? Explain their four categories with examples, state their importance, and name two related careers.
6. "Computing has both positive and negative impacts on society." Discuss with reference to education, healthcare and agriculture, and suggest three rules for smart and responsible use.

**Answer key.** Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

## Check yourself — answers

- 5.1 Introduction to Computing Impacts.** Answers: 1. Any four of: schools, hospitals, banks, farms, industries. 2. The effect computing has on people and society — it can be positive or negative.
- 5.1.1 Artificial Intelligence (AI).** Answers: 1. Because it is created by humans. 2. AI is the brain; IoT is the eyes. 3. Yes — it uses data (past spam emails) to decide whether a new email is spam, a task that needs "thinking".
- 5.1.2 Internet of Things (IoT) & Components.** Answers: 1. (a) sensor, (b) actuator, (c) sensor, (d) actuator. 2. Wi-Fi, Bluetooth, ZigBee, 4G/5G. 3. Because IoT devices produce massive amounts of data; the cloud stores and processes it over the internet instead of on small local hard drives.
- 5.1.3 Data Analytics & Components.** Answers: 1. Collection, Storage, Cleaning, Analysis, Visualization. 2. Duplicate records, incorrect values (typos), missing records, improper formats. 3. Data Visualization.
- Table: IoT vs AI vs Data Analytics Comparison.** Answers: 1. IoT. 2. Reports, graphs and insights.
- 5.2 Uses of IoT in Different Areas.** Answers: 1. They send alerts when full, so only full bins are emptied. 2. Rural areas lack doctors; devices send patient data directly to doctors far away. 3. The water pump (its switch).
- 5.2.1 Scope of AI in Education in Pakistan.** Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.
- 5.3 Information Sources (Primary, Secondary, Tertiary).** Answers: 1. P 2. S 3. T 4. P 5. S 6. T 7. P 8. S
- 5.4 Impacts of Computing in Various Fields.** Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked. 2. Poor internet access in rural areas. 3. Protect personal information/use strong passwords; do not share fake news; balance screen time with physical activity.
- 5.5 Assistive Technologies.** Answers: 1. Sensory aids. 2. Any two: audiobooks, spell-checkers, digital reminder apps. 3. Devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society.
- 5.5.1 Importance of Assistive Technologies.** Answers: 1. Equal access to education, independence in daily life, social inclusion. 2. They reduce the need for constant human assistance.
- 5.5.2 Career Connection in Assistive Tech.** Answers: 1. A biomedical engineer. 2. That websites are easy to navigate for visually impaired users.

## Answer key

### Multiple choice

- |      |      |      |       |       |
|------|------|------|-------|-------|
| 1. b | 4. d | 7. a | 10. b | 13. c |
| 2. c | 5. c | 8. b | 11. d | 14. d |
| 3. a | 6. d | 9. c | 12. a | 15. b |

### Short questions — model answers

- AI is human-made ("artificial") technology that can perform tasks requiring thinking ("intelligence"); it uses data to make decisions. Examples: face recognition to unlock a phone; YouTube suggesting videos; chatbots answering questions.

2. IoT is a system in which physical devices are connected to the internet and can collect and share data to perform tasks automatically. Example: a smartwatch collects heart-rate data and sends it to a mobile phone for display.
3. A sensor collects information from the physical world (input; the "eyes and ears"), e.g. temperature sensor, motion sensor. An actuator performs a physical action in response to a command, converting digital signals into movement (output), e.g. electric motor, automatic door opener.
4. Cloud computing stores IoT data and provides processing services over the internet instead of local hard drives, so it can handle the massive amounts of data produced by many devices. Examples: Google Drive, iCloud, online LMS.
5. Collection, storage, cleaning, analysis, visualization. Cleaning removes duplicated records, incorrect values (typos), missing files/records, and improper formats.
6. Smart cities: sensors adjust traffic signals to reduce congestion; smart bins send alerts when full (also street lighting, parking). Healthcare: smartwatches measure heart rate and vitals; remote monitoring devices send patient data to doctors (also emergency alerts).
7. Benefits: students learn at their own speed; AI identifies students who need extra help (also supports students with disabilities). Challenges: limited internet access in rural areas; lack of computers/devices (also teacher training, data privacy).
8. Primary sources give original, firsthand information from people, events or official records (e.g. interviews, NADRA records). Secondary sources contain information already collected and explained by someone else, summarising or analysing primary data (e.g. textbooks, news reports).
9. When the information is outdated, lacks proper references/citations, or has not been verified by an expert.
10. (a) Communication: + emails and video calls give instant, global communication at almost no cost; – less in-person interaction causes social isolation. (b) Business: + online shopping, digital banking and automation improve speed and efficiency; – manual jobs are lost to machines and banks face cyber-attacks.

### Long questions — model answers

1. **Definition:** IoT is a system in which physical devices are connected to the internet and collect and share data to perform tasks automatically. **Components:** (1) Sensors — collect information from the physical world, e.g. temperature, motion, fingerprint, smoke, soil moisture sensors. (2) Actuators — perform physical actions in response to commands, converting digital signals to movement, e.g. motor, switch, door opener, water valve, fan controller. (3) Connectivity — lets devices send/receive data, e.g. Wi-Fi, Bluetooth, ZigBee, 4G/5G. (4) Data processing — changes raw data into useful information through collecting, organising, analysing and deciding, e.g. checking if a room is too hot to switch on the AC. (5) Cloud computing — stores data and provides services over the internet, handles massive data, e.g. Google Drive, iCloud, LMS. (6) User interface — lets users see information and give commands, e.g. mobile apps, touch screens, voice commands. **Diagram:** Sensor (collects) → Wi-Fi → Cloud (processes) → Actuator (acts) → User App (UI).

2. **Table:** IoT — network of physical devices connected to the internet; purpose: data collection and automation; works with hardware; output: raw data from environment. AI — technology enabling machines to think and decide; purpose: intelligent decisions and predictions; works with smart software; output: smart decisions or actions. Data Analytics — process of examining data to find patterns; purpose: extracting meaningful insights; works with data (numbers, records); output: reports, graphs, insights. **Working together:** IoT is the eyes that gather data, Analytics sorts it, AI is the brain that decides. Example: a temperature sensor reads 36 °C (IoT) → recent readings are organised showing the room is heating up (Analytics) → AI decides the room is too hot (AI) → a switch turns on the AC (action).
3. (1) **Collection:** gather marks of all Maths tests, attendance and survey responses from students. (2) **Storage:** keep them safely in a spreadsheet or database, e.g. on the cloud. (3) **Cleaning:** remove duplicate records, correct typing errors (e.g. 120 out of 100), fill or mark missing records, and make formats (dates, roll numbers) consistent. (4) **Analysis:** study the clean data to find patterns — e.g. which chapters most students got wrong, whether low attendance links to failing. (5) **Visualization:** present results in bar charts or a pie chart of pass vs fail so the principal sees the problem at a glance and plans extra classes for weak chapters.
4. **Primary:** original, firsthand, highly authentic (interviews, photographs, diaries, NADRA records). **Secondary:** collected and explained by someone else, summarises/analyses primary data (textbooks, news reports, research papers, Bureau of Statistics reports). **Tertiary:** highly summarised and organised from primary and secondary sources for quick reference (encyclopedias, Wikipedia, dictionaries, handbooks). **Classification:** (i) secondary, (ii) primary, (iii) tertiary, (iv) primary, (v) secondary, (vi) tertiary. **Unreliability:** primary — incomplete, recorded incorrectly, or deliberately false; secondary — outdated, no references/citations, not verified by an expert; tertiary — unknown/unverified website, or oversimplified and missing key facts. Rule: do not trust an article without author, date or references.
5. **Definition:** devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society. **Categories:** (1) Mobility aids — help people move; smart wheelchairs, stair lifts. (2) Sensory aids — support vision and hearing; hearing aids, screen readers (VoiceOver), Braille displays. (3) Communication aids — help people with speech difficulties express themselves; speech-generating devices, text-to-speech apps. (4) Cognitive/learning aids — help with memory, reading, understanding; audiobooks, spell-checkers, reminder apps. **Importance:** equal access to education (screen readers let blind students read digital textbooks), independence in daily life (mobility aids reduce need for constant help), social inclusion (people work, study and socialise confidently). **Careers:** any two of assistive software developer (speech-to-text tools), biomedical engineer (artificial limbs, hearing aids), UX/UI accessibility expert (websites usable by visually impaired users).
6. **Introduction:** computing makes life faster and more connected but brings privacy risks and addiction. **Education:** + online classes, e-books and learning apps make study flexible and accessible; – excessive screen time reduces attention span and harms physical

health. **Healthcare:** + digital patient records and telemedicine let doctors treat remote patients accurately (e.g. rural Sindh); – patient data privacy is at risk if hospital servers are hacked. **Agriculture:** + smart irrigation and weather forecasting greatly increase yields; – poor rural internet means many farmers cannot use these tools. **Responsible use:** protect personal information and use strong passwords; do not share fake news or unverified forwards; balance screen time with sports and physical activity. **Conclusion:** benefits are large if technology is used wisely.

### Self-assessment: I can...

- ☐ define AI and explain why it is called "artificial intelligence".
- ☐ define IoT and name its six components with examples.
- ☐ list the five components of Data Analytics in order and describe data cleaning.
- ☐ describe IoT uses in smart cities, industry, healthcare, education, homes and farms.
- ☐ classify any source as primary, secondary or tertiary and judge its reliability.
- ☐ explain the four categories of assistive technologies and their importance.
- ☐ explain how IoT, Data Analytics and AI work together (eyes → sort → brain → action).
- ☐ tell the difference between a sensor and an actuator.
- ☐ compare IoT, AI and Data Analytics in a table.
- ☐ state uses, benefits and challenges of AI in education in Pakistan.
- ☐ discuss positive and negative impacts of computing in five fields.
- ☐ name careers linked to IoT, AI and assistive technology and the skills they need.